

开关选型

计算公式

三相电流

$$I_{\max} = \frac{P}{\sqrt{3}U} (1 - X\%)$$

Delta:

$$I_{\max} = \frac{P}{3U} (1 - X\%)$$

P : Rated power of transformer (VA)

I : 额定相电流 (A)

S : 变压器额定容量 Rated Capacity (VA)

Ust : Step voltage (V)

确认级电压

Star接法:

$$U_{\text{st}} = \frac{U \times n\%}{\sqrt{3}}$$

Delta接法:

$$U_{\text{st}} = U \times n\%$$

U : Line voltage on tap winding (V)

I_{max} : Maximum through current of OLTC (A)

Ust : Step voltage (V)

X% : Maximum tap percentage used for current reduction

n% : Percentage per step

具体步骤

算出最大电流和级电压，参照这里

Type	CV2 III 350			CV2 III 600		
Max. Rated Through Current (A)	350			600		
Phase	I	III	III	I	III	III
Connection	-	Y	D	-	Y	D
Max. Rated Step Voltage (V)	2000 for 10 contacts 1500 for 12 contact			2000 for 10 contacts 1500 for 12 contact		
Rated Step Capacity (kVA)	700			800		

然后看OS里的绝缘强度，对照参数表是否符合型号

ON-LOAD TAP CHANGER DATA							
Tap changer designation	Type	Number of phases	Max. rated through-current (A)	Y / D	Um (kV)	Tap selector size (not suitable for CV and CV2)	Basic connection diagram
	CV2	III	350	Y	126		10191G
Designation of operating positions	Max.number of eff. turns.Position (17), Mid.Position (s) (9), Min.number of eff. Turns Position (1)						
Flange for mounting	☒ Standard mounting flange ☐ Bell-type mounting						
	☐ Special flange for replacement						
Insulating strength			Rated separate source AC withstand voltage (kV/50Hz 1min)		Rated lightning impulse withstand voltage (kV 1.2/50us)		
At the tap changer to ground			230		550		
Across regulation winding for the same phase (a)			50		200		
Between any selected and preselected taps of the tap selector (a ₁)			50		200		
Fine tap winding between phases (b)							
Across coarse and fine tap winding (c ₁)			100		400		
Coarse and fine winding between phases (c ₂)							
Across coarse tap winding (d)			230		550		